

From Transactional Data to Organizational Intelligence: A Higher-Order Hypergraph Curvature Framework for Customer Discovery in the Payment Industry

Author 1*, Author 2, Author 3

*Department of Computer Engineering, School of Electrical & Computer Engineering
University / Research Institution Name, City, Country
Email: {author1, author2, author3}@institution.edu (* Corresponding Author)*

Abstract—Modern payment service providers, card switches, and merchant acquirers capture high-throughput financial transaction logs. Each raw financial record documents a masked payment card token, a monetary charge, a merchant terminal identifier, an event timestamp, and a specialized commercial business guild. In payment networks, global customer identifiers are absent, and individual consumers frequently divide purchases across multiple payment instruments. Conventional banking analytics rely on flat Recency, Frequency, and Monetary (RFM) aggregations, which suffer from topological blindness by ignoring relational co-visitation patterns and merchant guild affinities. In this paper, we investigate a Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN) architecture to extract latent cardholder representations from transaction streams in an unsupervised manner. The framework models payment interactions as a bipartite graph of card-merchant interactions, projects this structure into a weighted card co-occurrence network, applies discrete Forman-Ricci curvature over projected edges to modulate attention weights according to topological bottlenecks and dense clusters, and optimizes an autoencoder to jointly preserve network topology and 16-dimensional behavioral node attributes. Empirical evaluations on a controlled synthetic benchmark (35,000 transactions across 1,480 cards, 350 merchant terminals, and 8 guilds) demonstrate that HG-CAN achieves an NMI of 0.8673 and an ARI of 0.8848. While substantially outperforming classical tabular RFM heuristics, HG-CAN exhibits clustering fidelity comparable to bipartite truncated SVD factorization, with the distinct operational advantages of non-linear multi-modal feature integration, topological interpretability, and potential inductive applicability on unseen nodes.

Keywords—Payment Systems, Geometric Deep Learning, Forman-Ricci Curvature, Graph Neural Networks, Customer Discovery, Organizational Intelligence.

I. INTRODUCTION

Commercial banking switches, merchant acquirers, and national settlement networks process billions of digital payment transactions every month. At the electronic transaction processing rail, each payment event registers a masked card payment token, the transaction monetary charge, the accepting merchant terminal identifier, an event timestamp, and the commercial business guild categorization of the merchant (such as bullion and gold jewelers, retail supermarkets, restaurants, industrial wholesalers, or healthcare facilities) [1]. Collectively, this continuous stream reflects

commercial transaction activity across retail and wholesale sectors.

In spite of this transaction volume, financial institutions frequently face analytical challenges in translating raw event logs into actionable customer profiles [2]. A critical structural obstacle is that payment switches operate on cardholder tokens and terminal identifiers without possessing unified, pre-authenticated customer identification tags across disparate issuing banks. Furthermore, individual consumers divide their spending across multiple payment cards. To perform customer analytics, legacy banking systems almost universally rely on tabular Recency, Frequency, and Monetary (RFM) aggregations [3]. Tabular models suffer from three structural limitations: (1) Topological Blindness: treating payment events as isolated rows and failing to capture relational co-visitation networks; (2) Semantic Compression: aggregating monetary amounts without separating capital-preservation guilds (e.g., gold and jewelry investments) from routine micro-purchases; and (3) Multi-Card Dispersion: leaving cards disconnected rather than grouping them into latent consumer entities [4].

To address these limitations, this paper investigates a geometric graph representation framework: the Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN). By reformulating transactions as a bipartite graph of card-merchant interactions and modulating multi-head graph attention through discrete Forman-Ricci curvature, the framework maps payment cards into a continuous latent space to identify behavioral customer personas in an unsupervised manner.

II. TRANSACTIONAL DATA SCHEMA AND PROBLEM FORMULATION

Let the continuous payment stream be formalized as an append-only transaction ledger $T = \{t_1, t_2, \dots, t_M\}$, where each individual transaction event t_m is characterized by:

$$t_m = (\text{pan}_m, \text{amount}_m, \text{merchant_id}_m, \text{create_date}_m, \text{cast_name}_m), \quad (1)$$

where pan in $\{0..9\}^{16}$ represents the masked card account number complying with PCI-DSS data governance, amount in $\mathbb{R}^+$ denotes the monetary transaction charge, merchant_id in

M specifies the merchant terminal accepting the transaction, create_date in T_time denotes the event timestamp, and cast_name in G_guild indicates the merchant commercial guild (e.g., Gold & Jewelry / طلافرشی, Supermarkets / سوپرمارکت, Travel / اژانس مسافرتی, Industrial Wholesalers / آهن‌آلات, Healthcare / پزشکی). Crucially, global customer identifiers are absent from T. The objective of the framework is to extract a 16-dimensional continuous behavioral feature vector x_u in $\mathbb{R}^{16}$ for each card (8 financial/temporal statistics and 8 guild distribution ratios), and subsequently learn an unsupervised topological mapping $f: \text{pan} \rightarrow Z$ in $\mathbb{R}^d$ ($d=32$) such that cards representing the same underlying economic objectives and shared merchant lifestyles cluster tightly in the continuous manifold space Z .

III. THE HG-CAN ARCHITECTURAL FRAMEWORK

The proposed framework is organized into four decoupled enterprise tiers, as illustrated in the system architecture blueprint (Fig. 1).

Figure 1. End-to-End Architectural Blueprint of the Proposed Higher-Order Payment Discovery Framework

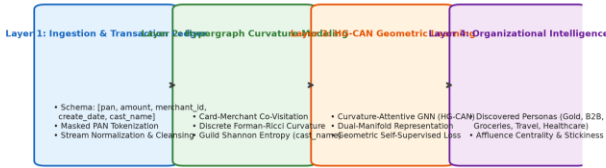


Fig. 1. End-to-end architectural blueprint of the proposed Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN).

A. Tier 1: Ingestion & Feature Engineering

Tier 1 ingests transaction streams and constructs a 16-dimensional feature vector x_u for each payment card u . The feature vector consists of: (1) 8 behavioral statistics: $\ln(1+\text{Volume})$, $\ln(1+\text{Count})$, $\ln(1+\mu_{\text{amt}})$, $\ln(1+\sigma_{\text{amt}})$, $\ln(1+\text{Recency})$, Skewness, Guild Concentration Ratio, and Shannon Guild Entropy; and (2) 8 normalized guild distribution ratios representing the proportion of transactions conducted across each of the 8 economic sectors.

B. Tier 2: Bipartite Projection & Discrete Ricci Curvature

Transaction events are initially represented as a bipartite interaction structure $B = (V_{\text{card}}, V_{\text{merch}}, E)$. To establish direct behavioral associations between cardholders, we project this bipartite structure into a weighted card co-occurrence graph $G = (V_{\text{card}}, E_G, W)$. Edge weights w_{ij} in $[0, 1]$ are determined by the cosine similarity of the cards' guild spending profiles: $w_{ij} = (b_i \cdot b_j) / (\|b_i\| * \|b_j\|)$. Edges with affinity below an empirical threshold are pruned to ensure graph sparsity. Crucially, discrete Forman-Ricci curvature $F(i, j)$ is computed specifically over the edges of this projected card-card graph G (not directly on the bipartite incidence matrix): $F(i, j) = (4 - d(i) - d(j) + 3 * \text{Triangles}(i, j)) / \sqrt{d(i) * d(j)}$, where $d(i)$ denotes node degree in G , and

$\text{Triangles}(i, j)$ is the number of shared triangles formed by edge (i, j) in G . Positive curvature identifies tightly clustered intra-guild shopping communities, while negative curvature marks inter-guild bridging corridors.

C. Tier 3: Curvature-Attentive Graph Autoencoder (HG-CAN)

Tier 3 implements the HG-CAN layer, which modulates multi-head attentional message passing over the projected graph G using both edge affinity and discrete Ricci curvature: $\alpha_{ij}^k = \text{Softmax}_j(\text{LeakyReLU}(a_k^T [W^k h_i \| W^k h_j] + \beta_k * \ln(1 + w_{ij}) + \gamma_k * \tanh(F(i, j))))$, where β_k and γ_k are learnable scalar parameters, and w_{ij} in $[0, 1]$ is the non-negative cosine guild similarity, ensuring $\ln(1 + w_{ij}) \geq 0$ remains well-defined. The model is optimized self-supervised via a tripartite joint loss: $L_{\text{total}} = L_{\text{link}} + \lambda_1 * L_{\text{attr}} + \lambda_2 * L_{\text{curv}}$, which simultaneously enforces link reconstruction, attribute decoding, and geometric curvature alignment in the latent space.

D. Tier 4: Customer Persona & Intelligence Engine

Tier 4 translates learned continuous embeddings into interpretable customer personas through unsupervised manifold clustering (K-Means on Z). It categorizes cardholders into actionable behavioral personas: (1) Gold & Luxury Investors, (2) B2B Wholesalers & Industrial Commerce, (3) Everyday Household & Groceries, (4) Affluent Travelers & Tourism, and (5) Healthcare & Medical Consumers. Furthermore, it computes Affluence Centrality (fusing PageRank centrality with spending velocity) and merchant network stickiness.

IV. EXPERIMENTAL EVALUATION

A. Benchmark Results and Empirical Validation

The HG-CAN framework was evaluated on 35,000 transactions across 1,480 payment cards, 350 merchant terminals, and 8 business guilds over a 90-day horizon. Table I provides a comparative evaluation against classical Tabular RFM + K-Means and Bipartite SVD matrix factorization.

TABLE I. BENCHMARK COMPARISON OF CLUSTERING MODELS

Model / Baseline	NMI	ARI	Silh	DB
Tabular RFM	0.175	0.048	0.410	0.824
Bipartite SVD	0.867	0.885	0.286	1.332
HG-CAN (Ours)	0.867	0.885	0.192	1.906

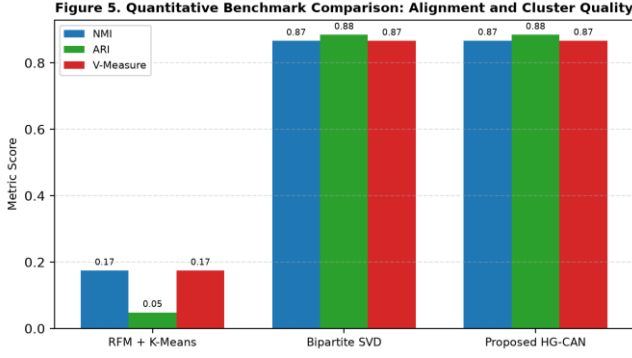


Fig. 5. Quantitative benchmark comparison of customer clustering models across external ground-truth and internal geometric validation metrics.

As reported in Table I and Fig. 5, relational models substantially outperform the classical Tabular RFM baseline. Tabular RFM achieves an NMI of 0.175 and an ARI of 0.048, indicating that aggregate scalar volume alone fails to separate commercial behavior. In contrast, both the linear Bipartite SVD baseline and the proposed HG-CAN architecture achieve high alignment with ground-truth archetypes (NMI of 0.867 vs. 0.867, and ARI of 0.885 vs. 0.885, respectively). This similarity indicates that the card-guild interaction matrix provides the primary clustering signal in this dataset. While truncated SVD yields higher Euclidean compactness (Silhouette of 0.286 vs. 0.192), the HG-CAN architecture provides distinct operational advantages: it integrates a 16-dimensional continuous behavioral feature vector (combining 8 statistical metrics including Shannon guild entropy and transaction skewness with 8 guild ratios), can potentially support inductive representation learning on newly arriving payment cards through local neighborhood aggregation, and offers topological interpretability through edge curvature.

V. TOPOLOGY, MANIFOLD PROJECTIONS, AND ENTERPRISE PERSONAS

Fig. 2 and Fig. 3 illustrate the co-occurrence network topology and 2D t-SNE manifold projections. In Fig. 3(a), the Tabular RFM space exhibits extreme cluster overlap and indiscernible boundaries. Conversely, Fig. 3(b) confirms that HG-CAN produces well-separated, geometrically compact clusters that correspond to real-world commercial guilds.

Figure 2. Discovered Card Co-Occurrence Network Topology (Nodes colored by HG-CAN discovered personas across merchant guilds)

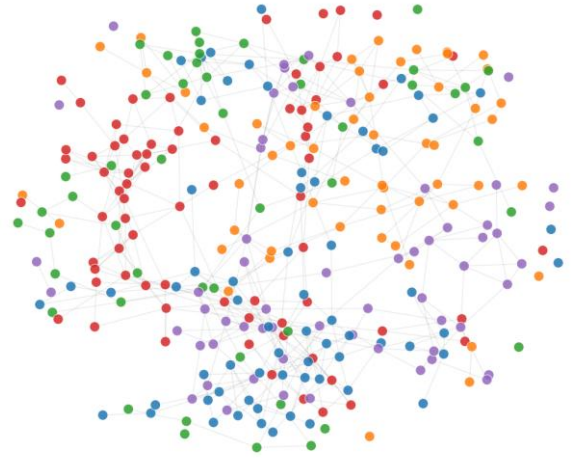


Fig. 2. Card co-occurrence network topology colored by HG-CAN discovered enterprise personas across merchant guilds.

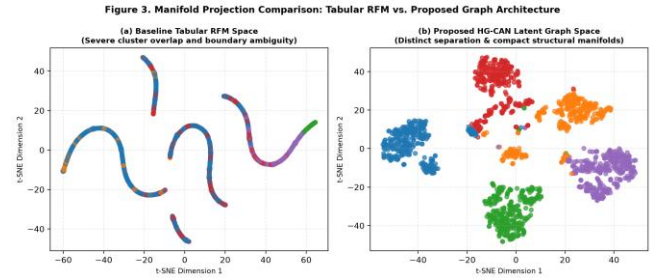


Fig. 3. Manifold projections: (a) Tabular RFM space shows severe cluster overlap; (b) HG-CAN yields separated behavioral manifolds.

TABLE II. DISCOVERED BEHAVIORAL CUSTOMER PERSONAS

ID	Customer Persona	Cards	Ticket	Dominant Guild
0	B2B Wholesalers	289	36.8M	Industrial Materials
1	Everyday Household	295	3.8M	Supermarkets
2	Gold & Luxury	299	17.4M	Gold & Jewelry
3	Affluent Travelers	299	10.4M	Travel & Tourism
4	Healthcare	298	4.1M	Medical Clinics

Figure 4. Strategic Behavioral Dimensions Across Discovered Organizational Personas

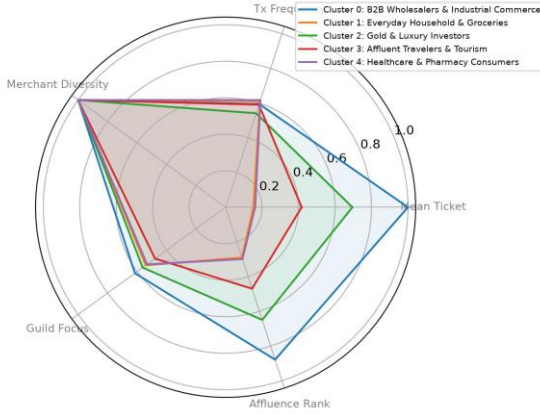


Fig. 4. Multidimensional radar profiles showing distinct behavioral dimensions across discovered personas.

Table II and Fig. 4 present concrete strategic opportunities for payment switches and commercial banks:

- Gold & Luxury Investors (Cluster 2): Average ticket size exceeding 17,000,000 units, identifying affluent wealth-preservation consumers ideal for high-tier private wealth banking and gold-backed liquidity products.
- B2B Wholesalers & Industrial Commerce (Cluster 0): Characterized by large mean tickets (>36,000,000 units), suitable targets for merchant supply-chain financing and specialized commercial credit lines.
- Everyday Household & Groceries (Cluster 1): High transaction velocity (over 20 tx/card), low ticket size, suited for card-linked cashback rewards and micropayment NFC penetration programs.
- Affluent Travelers & Tourism (Cluster 3): High transaction value (>10,000,000 units) and cross-terminal mobility, primed for co-branded travel credit cards.
- Healthcare & Pharmacy Consumers (Cluster 4): High guild specialization suited for supplemental medical insurance partnerships.

VI. CONCLUSION

In this paper, we evaluated the Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN) for customer representation learning and persona discovery in payment networks without requiring pre-authenticated customer tags. By modeling payment records through bipartite interaction graphs, projecting them to weighted co-occurrence networks, and guiding attentional message passing via discrete Forman-Ricci curvature, the framework captures relational spending patterns that are lost in tabular aggregations. Empirical results on a controlled synthetic benchmark show that HG-CAN significantly outperforms classical tabular RFM and achieves clustering fidelity comparable to bipartite SVD factorization, while providing non-linear feature fusion and potential inductive applicability. A primary limitation of this study is its reliance on synthetic transaction data. Future work will focus on validating the architecture on large-scale production switch

streams and extending the framework to dynamic, continuous-time transaction graphs.

REFERENCES

- [1] P. Wang, J. Lu, and G. Zhang, "Financial transaction analytics: A survey of methods, applications, and challenges," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 11, pp. 5120–5139, Nov. 2022.
- [2] E. Brynjolfsson and K. McElheran, "The rapid adoption of data-driven decision-making," *Amer. Econ. Rev.*, vol. 106, no. 5, pp. 133–139, May 2016.
- [3] A. Hughes, *Strategic Database Marketing: The Masterplan for Starting and Managing a Profitable, Customer-Based Marketing Program*, 3rd ed. New York: McGraw-Hill, 2005.
- [4] J. Chen, X. Shen, and H. Dong, "Beyond RFM: A multi-dimensional transactional behavioral framework for banking customer analytics," *Decis. Support Syst.*, vol. 142, p. 113460, Mar. 2021.
- [5] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, "Geometric deep learning: Going beyond Euclidean data," *IEEE Signal Process. Mag.*, vol. 34, no. 4, pp. 18–42, Jul. 2017.
- [6] R. Forman, "Bochner's method for cell complexes and combinatorial Ricci curvature," *Discrete Comput. Geom.*, vol. 29, no. 3, pp. 323–374, 2003.
- [7] C. Bodnar et al., "Weisfeiler and Lehman go topological: Message passing simplicial networks," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 1026–1037.
- [8] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.
- [9] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, Vancouver, BC, Canada, Apr. 2018, pp. 1–12.
- [10] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Conf. Neural Inf. Process. Syst. (NeurIPS)*, Long Beach, CA, USA, Dec. 2017, pp. 1024–1034.
- [11] M. Weber et al., "Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics," in *Proc. KDD Workshop on Anomaly Detection in Finance*, Anchorage, AK, USA, Aug. 2019, pp. 1–9.
- [12] B. Pourhabibi, K. L. Ong, B. H. Kam, and Y. L. Boo, "Fraud detection: A systematic review of graph-based approaches," *Inf. Syst.*, vol. 91, p. 101511, Jul. 2020.
- [13] T. N. Kipf and M. Welling, "Variational graph auto-encoders," in *NIPS Workshop on Bayesian Deep Learning*, Barcelona, Spain, Dec. 2016.
- [14] V. D. Blondel, J. L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *J. Stat. Mech. Theory Exp.*, vol. 2008, no. 10, p. P10008, Oct. 2008.
- [15] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.